# TRIMETAZIDINE

Parkinsonism risk

Add-on therapy only

**CHEMICAL NAME** 1-[(2,3,4-trimethoxyphenyl)methyl]piperazine (dihydrochloride)
**CLASS** Metabolic antianginal agent (3-KAT inhibitor)
**TRADE NAMES** Vastarel MR · Preductal MR · Trimetazidine (generic)

## THERAPEUTIC USES / INDICATIONS

- Stable angina; add-on in patients not adequately controlled by or intolerant of first-line drugs.
- (Previously used for vertigo/tinnitus — no longer recommended.)

## ADMINISTRATION

- **PO** MR tablet twice daily with meals (morning and evening).

## DOSAGES

- **Adult:** 35 mg MR twice daily.
- **Moderate renal impairment (CrCl 30–60) or elderly:** 35 mg once daily.

## PHARMACOKINETICS

- **Peak:** 5 h (MR). Bioavailability ~90%.
- Minimal metabolism.
- Excreted mainly unchanged in urine. **t½:** 7 h (longer in elderly/renal impairment).

## PHARMACODYNAMICS

- Inhibits mitochondrial **long-chain 3-ketoacyl-CoA thiolase** → shifts cardiac metabolism from fatty-acid to glucose oxidation (uses less oxygen per ATP).
- Anti-ischaemic without affecting heart rate or BP.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Dizziness, headache.
- Nausea, abdominal pain, dyspepsia.
- Fatigue, rash.

### ADVERSE EFFECTS (SERIOUS)

- Parkinsonian symptoms (tremor, rigidity, gait instability), restless legs — reversible on stopping.
- Falls in elderly.
- Thrombocytopenia, hepatitis (rare).

## CONTRAINDICATIONS & CAUTIONS

- Parkinson's disease, parkinsonism, tremor, restless legs, other movement disorders.
- Severe renal impairment (CrCl <30).
- Pregnancy and breastfeeding (lack of data).

### KEY INTERACTIONS

- No major interactions; caution with other drugs causing extrapyramidal effects.

## NURSING MANAGEMENT

- Assess for tremor, rigidity, unsteady gait; report.
- Check renal function for dosing.
- Fall precautions in elderly.
- Monitor angina frequency.

## PATIENT EDUCATION

- Take with breakfast and dinner.
- Report trembling, stiffness or problems walking.
- Keep using your other heart medicines and GTN.

## KEY POINTS

- Metabolic antianginal — add-on only.
- Causes parkinsonism; avoid in movement disorders.
- No effect on HR or BP.
- Renal dose adjustment.